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arXiv:2210.04797 (q-fin)

[Submitted on 23 Sep 2022 (v1), last revised 8 Aug 2024 (this version, v3)]

DeepVol: Volatility Forecasting from High-Frequency Data with Dilated Causal Convolutions

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Volatility forecasts play a central role among equity risk measures. Besides traditional statistical models, modern forecasting techniques based on machine learning can be employed when treating volatility as a univariate, daily time-series. Moreover, econometric studies have shown that increasing the number of daily observations with high-frequency intraday data helps to improve volatility predictions. In this work, we propose DeepVol, a model based on Dilated Causal Convolutions that uses high-frequency data to forecast day-ahead volatility. Our empirical findings demonstrate that dilated convolutional filters are highly effective at extracting relevant information from intraday financial time-series, proving that this architecture can effectively leverage predictive information present in high-frequency data that would

otherwise be lost if realised measures were precomputed. Simultaneously, dilated convolutional filters trained with intraday high-frequency data help us avoid the limitations of models that use daily data, such as model misspecification or manually designed handcrafted features, whose devise involves optimising the trade-off between accuracy and computational efficiency and makes models prone to lack of adaptation into changing circumstances. In our analysis, we use two years of intraday data from NASDAQ-100 to evaluate the performance of DeepVol. Our empirical results suggest that the proposed deep learning-based approach effectively learns global features from high-frequency data, resulting in more accurate predictions compared to traditional methodologies and producing more accurate risk measures.

Comments: Updated version
Subjects: **Risk Management (q-fin.RM)**; Machine Learning (cs.LG); Computational Finance (q-fin.CP)
Cite as: **arXiv:2210.04797 [q-fin.RM]**
(or **arXiv:2210.04797v3 [q-fin.RM]** for this version)
https://doi.org/10.48550/arXiv.2210.04797
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Submission history

From: Fernando Moreno-Pino [view email]
[v1] Fri, 23 Sep 2022 16:13:47 UTC (1,384 KB)
[v2] Thu, 13 Oct 2022 09:59:26 UTC (1,426 KB)
[v3] Thu, 8 Aug 2024 11:26:01 UTC (774 KB)

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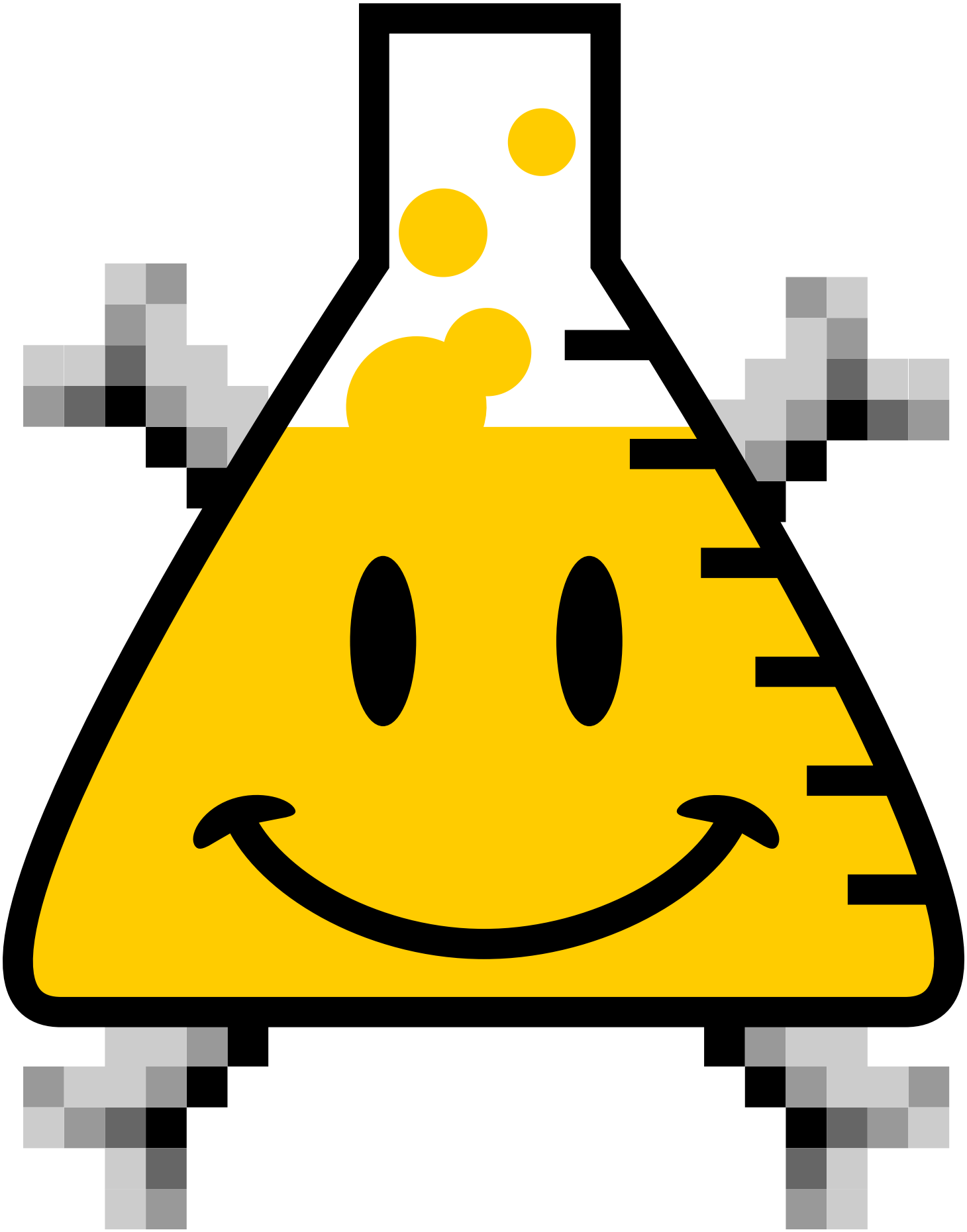

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